Supplementary material for

Effect Identity for LLM Agents: Schema-Determined Limits on Duplicate Suppression

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S1 What this document contains, and how to regenerate it

This supplement holds the material that the main paper summarises: the full configuration, ablation and per-category tables; the secondary experiments; the details of the label-validation study, including two defects in its preregistered analysis; the cost accounting; an errata list against the earlier long-form version of this work; and the reproduction instructions.

Every table here is generated from the artifact's results files by `tools/make-supp-tables.mjs`, and every interval quoted in this document and in the main paper is checked against those files by `checkPaper.js`. The whole pipeline is one command:

```
node docs/research/aij/build.mjs
```

which regenerates the figure data, checks every quoted interval and headline number, runs the artifact's verifier, and compiles both documents. A quoted number that no results file contains fails the build.

S2 Corpus construction

The corpus holds 1,066 labelled pairs over six irreversible tools; the main paper's Table 6 lists the categories. Labels are by construction. A duplicate (SAME) pair is generated by a transformation instructed to preserve the act, and a validator rejects any variant that alters a pinned numeric mention or a recipient; a distinct (DIFF) pair mutates a declared field, inserts a distinguishing token, or is hand-specified to differ in intent. The paraphrase categories are produced by four arms, three of them language models at temperature 0.9 and one written by hand with no model in the loop (Table S1). The corpus was grown once, additively, during an earlier revision: the loose-schema rewrite category from 15 to 75 pairs and the hand-specified semantic-difference category from 10 to 61, under a separate seed, with no existing pair modified. Both cells previously carried qualitative claims at 8 and 5 test pairs respectively.

Table S1: Paraphrase generator arms and the validator's rejections at generation time. The hand-written arm has no model in the loop. Mechanical and hand-specified pairs (650) have no generator.

Generator arm	pairs	cal	test	S2 kept/dropped	S4 kept/dropped
handwritten-authors	68	34	34	68/0	0/0
llama-3.3-70b-versatile	109	54	55	60/8	49/19
openai/gpt-oss-120b	121	60	61	59/9	62/6
qwen/qwen3.6-27b	118	59	59	66/2	52/6

S3 The parser defect, and why the artifact ships it

An earlier scoring pass over a 780-pair version of this corpus instructed the judge to answer with an integer percentage and parsed the reply with a permissive parser that accepted any finite number and divided by 100. A reply of `0.95` — a probability rather than a percentage — was therefore stored as `0.0095`, turning a confident duplicate verdict into a confident distinct score. The reply is valid JSON, so it never triggered a retry and never entered a failure count: the harness reported a 0% failure rate for a pass in which a substantial fraction of scores were inverted.

The contamination is identifiable exactly rather than estimated, because no integer percentage can produce a stored score that is not a multiple of 0.01: 180 of 780 scores (23.1%) are not. Every judge-dependent number from that pass was withdrawn. The affected corpus and scores are kept under `archive/` so that the defect and its extent remain auditable, and the strict contract described in the main paper (one accepted form, no rescaling, no clamping, abstention recorded as `null`) exists because of it. The lesson generalises beyond this artifact: an evaluation that instruments only whether an oracle *replied* cannot detect an oracle that replied in the wrong units.

S4 Judge coverage, admission and abstention

Table S2: Judge coverage and the admission rule (coverage $\geq 80\%$, at least two distinct score values, AUC ≥ 0.60). AUC is computed over all scored pairs. Failed clauses are abbreviated: *coverage* (too few pairs scored to calibrate or to measure), *resolution* (fewer than two distinct score values), *signal* (no scored SAME/DIFF overlap, so discrimination is unmeasurable). Exact permutation tests (20,000 permutations) find no association between the Qwen judge's abstentions and any stratification.

Judge	scored / 1,066	coverage	AUC	admitted
Llama-3.3-70B	60	5.6%	1.000	no (coverage)
GPT-OSS-120B	1066	100.0%	0.962	yes
GPT-OSS-20B	1066	100.0%	0.962	yes
Qwen3.6-27B	1041	97.7%	0.974	yes
Allam-2-7B	0	0.0%	—	no (coverage, resolution, signal)

Table S3: Abstention homogeneity for the Qwen judge (25 abstentions among 1066 pairs), by exact permutation test.

Stratification	levels	χ^2	permutation p
category	9	11.14	0.183
label	2	0.59	0.547
tool	6	9.37	0.091
split	3	1.93	0.363
generator	5	3.66	0.448

The admission rule was fixed before scoring: coverage of at least 80% of pairs, at least two distinct score values, and an AUC of at least 0.60. The second and third clauses exist because coverage alone would admit a judge that answers every pair identically: its calibration pool is a pile of one value, its threshold equals that value, a strict inequality then suppresses nothing, and the resulting table row reads "0.0% wrong suppression, 0.0% deduplication" — the appearance of a flawless judge manufactured by the absence of a judge. Two models fail admission on this corpus.

`llama-3.3-70b-versatile` returns a parseable score for 60 of 1,066 pairs; `allam-2-7b` returns none. The second was also run against an extension corpus of 1,043 pairs, where it scored 74 and answered 0.90 on 63 of them, so its exclusion is not only a coverage artefact: at that resolution no threshold separates anything.

S5 Full configuration tables

Table S4 reports every configuration behind the main paper's headline table: three judges, two levels, and both calibration pools. The rightmost column conditions on the residual, which is the population Theorem 1 is about; the difference between the two pools is the jurisdiction-matching effect discussed in the main paper.

Table S4: Every reported configuration: each judge at both levels, calibrated on its own jurisdiction (residual) and, for comparison, on the wider non-canonical pool. Rates are on the test split; the residual column conditions on the pairs the deployed rule routes to the judge.

Judge	α	pool n	t_{hi}	wrong suppression	dedup	silent escape	WS over residual
GPT-OSS-120B	0.01 (wider)	230	0.90	0.0% [0.0–1.6]	69.0% [63.4–74.2]	13.2% [9.7–17.6]	0.0% [0.0–2.6]
GPT-OSS-120B	0.01 (residual)	142	0.95	0.0% [0.0–1.6]	22.1% [17.6–27.3]	13.2% [9.7–17.6]	0.0% [0.0–2.6]
GPT-OSS-120B	0.05 (wider)	230	0.20	3.0% [1.5–6.1]	100.0% [98.7–100.0]	0.0% [0.0–1.3]	4.9% [2.4–9.7]
GPT-OSS-120B	0.05 (residual)	142	0.80	1.7% [0.7–4.3]	94.7% [91.4–96.7]	5.3% [3.3–8.6]	2.8% [1.1–6.9]
GPT-OSS-20B	0.01 (wider)	230	1.00	0.0% [0.0–1.6]	12.1% [8.8–16.4]	2.1% [1.0–4.6]	0.0% [0.0–2.6]
GPT-OSS-20B	0.01 (residual)	142	1.00	0.0% [0.0–1.6]	12.1% [8.8–16.4]	2.1% [1.0–4.6]	0.0% [0.0–2.6]
GPT-OSS-20B	0.05 (wider)	230	0.10	3.0% [1.5–6.1]	97.5% [94.9–98.8]	2.1% [1.0–4.6]	4.9% [2.4–9.7]
GPT-OSS-20B	0.05 (residual)	142	0.95	1.3% [0.4–3.7]	82.6% [77.7–86.6]	2.1% [1.0–4.6]	2.1% [0.7–5.9]
Qwen3.6-27B	0.01 (wider)	227	1.00	0.0% [0.0–1.6]	12.1% [8.8–16.4]	8.5% [5.8–12.4]	0.0% [0.0–2.6]
Qwen3.6-27B	0.01 (residual)	139	1.00	0.0% [0.0–1.6]	12.1% [8.8–16.4]	8.5% [5.8–12.4]	0.0% [0.0–2.6]
Qwen3.6-27B	0.05 (wider)	227	0.95	0.9% [0.2–3.1]	89.0% [84.8–92.1]	8.5% [5.8–12.4]	1.4% [0.4–4.9]
Qwen3.6-27B	0.05 (residual)	139	0.95	0.9% [0.2–3.1]	89.0% [84.8–92.1]	8.5% [5.8–12.4]	1.4% [0.4–4.9]

Table S5: Common-subset comparison: the 1041 pairs all three judges scored (499 calibration, 499 test, 43 attack), each judge calibrated on the common calibration split.

Judge	t_{hi}	wrong suppression	dedup	silent escape	flagged escape
GPT-OSS-120B	0.80	1.8% [0.7–4.4]	94.5% [91.1–96.6]	5.5% [3.4–8.9]	0.0% [0.0–1.4]
GPT-OSS-20B	0.95	1.3% [0.5–3.8]	82.0% [77.0–86.1]	2.2% [1.0–4.7]	15.8% [12.0–20.6]
Qwen3.6-27B	0.95	0.9% [0.2–3.2]	91.2% [87.2–94.0]	8.8% [6.0–12.8]	0.0% [0.0–1.4]

S6 Per-category decisions

S7 Baseline and ablation matrix

Two rows are worth singling out. The certifier contributes nothing to the *rates* on this corpus: removing it leaves every decision unchanged on the primary judge, because the judge re-derives what canonicalisation already knew. Its contribution is that the decisions it settles carry zero wrong suppression by construction and cost no model call. And no wrong-suppression comparison in the matrix is statistically significant: every one rests on at most four discordant items. We therefore claim no benign wrong-suppression advantage for any configuration over any other, and rest the safety case on the bound itself and on the injection class the refuter closes by construction.

Table S6: Per-category decisions of the proposed configuration on the test split, $\alpha = 0.05$, for each judge.

Judge	Category	n	suppress	escalate	execute
GPT-OSS-120B	D1 near-miss mutation	126	0.8% [0.1–4.4]	0.0% [0.0–3.0]	99.2% [95.6–99.9]
GPT-OSS-120B	D2 legitimate repeats	75	1.3% [0.2–7.2]	0.0% [0.0–4.9]	98.7% [92.8–99.8]
GPT-OSS-120B	D3 semantic difference	31	6.5% [1.8–20.7]	0.0% [0.0–11.0]	93.5% [79.3–98.2]
GPT-OSS-120B	S1 mechanical reformat	34	100.0% [89.8–100.0]	0.0% [0.0–10.2]	0.0% [0.0–10.2]
GPT-OSS-120B	S2 paraphrase	127	92.9% [87.1–96.2]	0.0% [0.0–2.9]	7.1% [3.8–12.9]
GPT-OSS-120B	S3 loose-schema value	38	100.0% [90.8–100.0]	0.0% [0.0–9.2]	0.0% [0.0–9.2]
GPT-OSS-120B	S4 adversarial paraphrase	82	92.7% [84.9–96.6]	0.0% [0.0–4.5]	7.3% [3.4–15.1]
GPT-OSS-20B	D1 near-miss mutation	126	0.0% [0.0–3.0]	2.4% [0.8–6.8]	97.6% [93.2–99.2]
GPT-OSS-20B	D2 legitimate repeats	75	4.0% [1.4–11.1]	16.0% [9.4–25.9]	80.0% [69.6–87.5]
GPT-OSS-20B	D3 semantic difference	31	0.0% [0.0–11.0]	12.9% [5.1–28.9]	87.1% [71.1–94.9]
GPT-OSS-20B	S1 mechanical reformat	34	100.0% [89.8–100.0]	0.0% [0.0–10.2]	0.0% [0.0–10.2]
GPT-OSS-20B	S2 paraphrase	127	81.9% [74.3–87.6]	16.5% [11.1–24.0]	1.6% [0.4–5.6]
GPT-OSS-20B	S3 loose-schema value	38	81.6% [66.6–90.8]	13.2% [5.8–27.3]	5.3% [1.5–17.3]
GPT-OSS-20B	S4 adversarial paraphrase	82	76.8% [66.6–84.6]	20.7% [13.4–30.7]	2.4% [0.7–8.5]
Qwen3.6-27B	D1 near-miss mutation	126	0.0% [0.0–3.0]	0.0% [0.0–3.0]	100.0% [97.0–100.0]
Qwen3.6-27B	D2 legitimate repeats	75	2.7% [0.7–9.2]	4.0% [1.4–11.1]	93.3% [85.3–97.1]
Qwen3.6-27B	D3 semantic difference	31	0.0% [0.0–11.0]	0.0% [0.0–11.0]	100.0% [89.0–100.0]
Qwen3.6-27B	S1 mechanical reformat	34	100.0% [89.8–100.0]	0.0% [0.0–10.2]	0.0% [0.0–10.2]
Qwen3.6-27B	S2 paraphrase	127	87.4% [80.5–92.1]	2.4% [0.8–6.7]	10.2% [6.1–16.7]
Qwen3.6-27B	S3 loose-schema value	38	97.4% [86.5–99.5]	0.0% [0.0–9.2]	2.6% [0.5–13.5]
Qwen3.6-27B	S4 adversarial paraphrase	82	82.9% [73.4–89.5]	4.9% [1.9–11.9]	12.2% [6.8–21.0]

Table S7: Baseline and ablation matrix (primary judge, $\alpha = 0.05$). Each row is calibrated on its own jurisdiction.

Configuration	pool n	t_{hi}	wrong suppression	dedup	silent escape
B0_no_fence	230	$+\infty$	0.0% [0.0–1.6]	0.0% [0.0–1.3]	100.0% [98.7–100.0]
B1_exact_string	0	$+\infty$	0.0% [0.0–1.6]	0.0% [0.0–1.3]	100.0% [98.7–100.0]
B2_content_addressed	230	$+\infty$	0.0% [0.0–1.6]	12.1% [8.8–16.4]	87.9% [83.6–91.2]
B3_raw_judge	230	0.50	2.2% [0.9–4.9]	99.6% [98.0–99.9]	0.4% [0.1–2.0]
B4_calibrated_judge	230	0.20	3.4% [1.8–6.7]	100.0% [98.7–100.0]	0.0% [0.0–1.3]
B5_certifier_judge	230	0.20	3.4% [1.8–6.7]	100.0% [98.7–100.0]	0.0% [0.0–1.3]
B6_refuter_judge	142	0.80	1.7% [0.7–4.3]	94.7% [91.4–96.7]	5.3% [3.3–8.6]
B7_decidable_only	142	$+\infty$	0.0% [0.0–1.6]	12.1% [8.8–16.4]	87.9% [83.6–91.2]
B8_full	142	0.80	1.7% [0.7–4.3]	94.7% [91.4–96.7]	5.3% [3.3–8.6]
B9_full_no_band	142	0.80	1.7% [0.7–4.3]	94.7% [91.4–96.7]	5.3% [3.3–8.6]
B10_full_no_calibration	142	0.50	2.2% [0.9–4.9]	99.6% [98.0–99.9]	0.4% [0.1–2.0]
B11_naive_multiplicity_m1	142	0.80	1.7% [0.7–4.3]	94.7% [91.4–96.7]	5.3% [3.3–8.6]
B12_corrected_multiplicity_m1	142	0.80	1.7% [0.7–4.3]	94.7% [91.4–96.7]	5.3% [3.3–8.6]
B11_naive_multiplicity_m5	142	0.80	1.7% [0.7–4.3]	94.7% [91.4–96.7]	5.3% [3.3–8.6]
B12_corrected_multiplicity_m5	142	0.95	0.0% [0.0–1.6]	22.1% [17.6–27.3]	13.2% [9.7–17.6]

Table S8: Exact McNemar tests over discordant items, Holm-corrected within family (liveness and safety corrected separately).

Family	Comparison	discordant	exact p	Holm
dedup	B8_full vs B2_content_addressed	232	2.9e-70	2.3e-69
dedup	B8_full vs B5_certifier_judge	15	6.1e-5	3.7e-4
dedup	B8_full vs B7_decidable_only	232	2.9e-70	2.3e-69
dedup	B8_full vs B3_raw_judge	14	1.2e-4	4.9e-4
dedup	B8_full vs B4_calibrated_judge	15	6.1e-5	3.7e-4
dedup	B8_full vs B6_refuter_judge	0	1.0e+0	1.0e+0
dedup	B8_full vs B9_full_no_band	0	1.0e+0	1.0e+0
dedup	B8_full vs B10_full_no_calibration	14	1.2e-4	4.9e-4
WS	B8_full vs B3_raw_judge	1	1.000	1.000
WS	B8_full vs B4_calibrated_judge	4	0.125	0.625
WS	B8_full vs B5_certifier_judge	4	0.125	0.625
WS	B8_full vs B10_full_no_calibration	1	1.000	1.000
WS	B8_full vs B12_corrected_multiplicity_m5	4	0.125	0.625

S8 Score distributions, the escalation band, and per-tool separation

Table S9: Score distribution and escalation band, $\alpha = 0.05$. "At extremes" counts scores ≤ 0.02 or ≥ 0.98 among non-certified pairs. Ties are residual duplicate test pairs whose score equals t_{hi} exactly; a strict inequality never suppresses them.

Judge	t_{hi}	t_{lo}	distinct values	SAME at extremes	DIFF at extremes	suppressed	tied at t_{hi}
GPT-OSS-120B	0.80	0.80	24	9.8% [7.5–12.7]	4.1% [2.7–6.2]	232/247	10
GPT-OSS-20B	0.95	0.00	6	83.7% [80.2–86.7]	87.2% [84.0–89.8]	198/247	40
Qwen3.6-27B	0.95	0.95	8	88.9% [85.8–91.4]	89.5% [86.5–91.9]	216/240	24

The band exists only when $t_{\text{lo}} < t_{\text{hi}}$. On the primary judge and on Qwen the two coincide, so the band is empty and a duplicate that fails to clear t_{hi} executes silently; on GPT-OSS-20B the band is wide and catches 15.3% of duplicates. What decides the outcome is mostly *ties*: judge scores are integer percentages, the suppression inequality is strict, and a duplicate whose score equals t_{hi} is never suppressed. Of the 20B's 43 banded duplicates, 40 score exactly at its threshold; on the primary judge 10 of 15 silent escapes and on Qwen all 24 are such ties. A deployment on a coarse score grid should expect the band's behaviour to be governed by this rather than by graded confidence.

S9 Per-tool calibration, ensembles and non-LLM baselines

Per-tool calibration is an axis trade rather than an improvement: splitting one pool six ways leaves two tools below the 19 distinct pairs needed to certify $\alpha = 0.05$, so those tools stop suppressing semantically altogether, which lowers wrong suppression and deduplication together. The non-LLM baselines are the clearest demonstration that the residual is not a surface-similarity problem: on the residual their AUC is *below* 0.5, because the distinct pairs that survive the refuter are textually near-identical while the duplicates are paraphrases.

Table S10: Per-tool score separation on residual test pairs. "Separable" means max over distinct pairs < min over duplicate pairs, so a single threshold achieves zero error in both directions on that tool.

Judge	Tool	n SAME	n DIFF	max DIFF	min SAME	separable
GPT-OSS-120B	charge_card	70	53	0.30	0.40	yes
GPT-OSS-120B	post_refund	31	20	0.90	0.85	no
GPT-OSS-120B	transfer_funds	40	20	0.90	0.70	no
GPT-OSS-120B	send_email	34	13	0.15	0.60	yes
GPT-OSS-120B	create_ticket	12	12	0.10	0.80	yes
GPT-OSS-120B	wire_money	60	26	0.10	0.75	yes
GPT-OSS-20B	charge_card	70	53	1.00	0.95	no
GPT-OSS-20B	post_refund	31	20	0.95	0.95	no
GPT-OSS-20B	transfer_funds	40	20	1.00	0.10	no
GPT-OSS-20B	send_email	34	13	0.05	0.00	no
GPT-OSS-20B	create_ticket	12	12	0.05	0.95	yes
GPT-OSS-20B	wire_money	60	26	0.05	0.00	no
Qwen3.6-27B	charge_card	67	51	1.00	0.95	no
Qwen3.6-27B	post_refund	30	20	1.00	0.95	no
Qwen3.6-27B	transfer_funds	38	19	0.05	0.95	yes
Qwen3.6-27B	send_email	33	13	0.95	0.95	no
Qwen3.6-27B	create_ticket	12	12	0.95	0.95	no
Qwen3.6-27B	wire_money	60	26	0.05	0.95	yes

Table S11: Per-tool calibration under the jurisdiction-matched pool. Tools with fewer than 19 residual distinct calibration pairs cannot certify $\alpha = 0.05$ and receive $t_{\text{hi}} = +\infty$ (no semantic suppression).

Judge	Tool	pool n	t_{hi}	certifiable at $\alpha = 0.05$
GPT-OSS-120B	charge_card	38	0.20	yes
GPT-OSS-120B	post_refund	28	0.90	yes
GPT-OSS-120B	transfer_funds	28	0.95	yes
GPT-OSS-120B	send_email	13	$+\infty$	no
GPT-OSS-120B	create_ticket	6	$+\infty$	no
GPT-OSS-120B	wire_money	29	0.80	yes
GPT-OSS-20B	charge_card	38	1.00	yes
GPT-OSS-20B	post_refund	28	1.00	yes
GPT-OSS-20B	transfer_funds	28	1.00	yes
GPT-OSS-20B	send_email	13	$+\infty$	no
GPT-OSS-20B	create_ticket	6	$+\infty$	no
GPT-OSS-20B	wire_money	29	0.10	yes
Qwen3.6-27B	charge_card	37	0.05	yes
Qwen3.6-27B	post_refund	28	1.00	yes
Qwen3.6-27B	transfer_funds	26	1.00	yes
Qwen3.6-27B	send_email	13	$+\infty$	no
Qwen3.6-27B	create_ticket	6	$+\infty$	no
Qwen3.6-27B	wire_money	29	0.95	yes

Table S12: Ensembles and the deterministic lexical scorer, calibrated on the jurisdiction-matched pool, $\alpha = 0.05$, with their injection rates.

Scorer	t_{hi}	wrong suppression	dedup	silent escape	D4a	D4b
ensemble_min	0.00	0.9% [0.2–3.1]	97.9% [95.4–99.0]	2.1% [1.0–4.6]	0.0% [0.0–16.1]	28.0% [14.3–47.6]
ensemble_mean	0.38	2.6% [1.2–5.5]	100.0% [98.7–100.0]	0.0% [0.0–1.3]	0.0% [0.0–16.1]	44.0% [26.7–62.9]
lexical	0.91	1.3% [0.4–3.7]	30.6% [25.5–36.2]	3.2% [1.7–6.0]	0.0% [0.0–16.1]	0.0% [0.0–13.3]

Table S13: Non-LLM similarity baselines through the identical conformal pipeline. An AUC below 0.5 means the score orders duplicates *below* distinct pairs on the residual: surface similarity is anti-correlated with identity there.

Scorer	residual AUC	conformal τ	residual dedup	system dedup
Token Jaccard	0.232	0.966	4.0%	15.7%
Char 3-gram Cosine	0.144	0.988	0.0%	12.1%
Levenshtein Ratio	0.101	0.995	0.0%	12.1%

Table S14: Multiple committed candidates, live judge (llama-3.1-8b-instant), $K = 50$ nested candidates, 40 distinct and 20 duplicate units. Wrong suppression is over the 20 held-out distinct units.

m	rule	t_{hi}	wrong suppression	dedup
1	uncorrected	0.90	0.0% [0.0–16.1]	50.0% [29.9–70.1]
1	bonferroni	0.90	0.0% [0.0–16.1]	50.0% [29.9–70.1]
1	setconformal	0.90	0.0% [0.0–16.1]	50.0% [29.9–70.1]
2	uncorrected	0.90	0.0% [0.0–16.1]	50.0% [29.9–70.1]
2	bonferroni	0.91	0.0% [0.0–16.1]	50.0% [29.9–70.1]
2	setconformal	0.91	0.0% [0.0–16.1]	50.0% [29.9–70.1]
5	uncorrected	0.90	0.0% [0.0–16.1]	50.0% [29.9–70.1]
5	bonferroni	0.91	0.0% [0.0–16.1]	50.0% [29.9–70.1]
5	setconformal	0.91	0.0% [0.0–16.1]	50.0% [29.9–70.1]
10	uncorrected	0.90	0.0% [0.0–16.1]	50.0% [29.9–70.1]
10	bonferroni	0.91	0.0% [0.0–16.1]	50.0% [29.9–70.1]
10	setconformal	0.91	0.0% [0.0–16.1]	50.0% [29.9–70.1]
25	uncorrected	0.90	0.0% [0.0–16.1]	50.0% [29.9–70.1]
25	bonferroni	$+\infty$	0.0% [0.0–16.1]	0.0% [0.0–16.1]
25	setconformal	$+\infty$	0.0% [0.0–16.1]	0.0% [0.0–16.1]
50	uncorrected	0.90	65.0% [43.3–81.9]	70.0% [48.1–85.5]
50	bonferroni	$+\infty$	0.0% [0.0–16.1]	0.0% [0.0–16.1]
50	setconformal	$+\infty$	0.0% [0.0–16.1]	0.0% [0.0–16.1]

S10 Multiple committed candidates

The live judge in this experiment assigns only two distinct values to its 3,000 comparisons (2,968 zeros and 32 ones). That saturation is what makes the experiment informative about multiplicity rather than about accuracy: every threshold in $[0, 1)$ induces the same decisions, so the result is a statement about m and not about the calibrated value. The intermediate points of the m-curve depend on the order in which candidates are appended to each unit and are not reported as a trend; the endpoint at $m = 50$ is a property of the full candidate set.

S11 What α promises: resampling and seed-level splits

Table S15: Realised wrong suppression over 300 resampled calibration pools per size, evaluated on the fixed residual test pairs (primary judge, $\alpha = 0.05$). The mean is flat in n; the tail and the exceedance share are not.

pool n	$P(t_{\text{hi}} = +\infty)$	mean	median	p95	max	share above α
10	100%	—	—	—	—	—
20	0%	2.8%	1.4%	13.2%	13.2%	9.7%
30	0%	1.8%	1.4%	4.9%	13.2%	3.7%
50	0%	2.1%	1.4%	4.9%	7.6%	1.3%
100	0%	2.9%	2.8%	4.9%	13.2%	2.0%

Table S16: Seed-level resplitting: 200 splits that partition the 219 seed acts first, so no seed appears on both sides. The shipped pair-level split is shown for comparison.

Judge	pair-split WS	seed-split mean	median	p95	share above α
GPT-OSS-120B	1.7% [0.7–4.3]	2.7%	2.8%	5.2%	10.5%
GPT-OSS-20B	1.3% [0.4–3.7]	2.2%	1.7%	4.2%	1.0%
Qwen3.6-27B	0.9% [0.2–3.1]	1.8%	1.3%	5.7%	11.0%

S12 Threshold sensitivity and judge degradation

Table S17: Threshold sweep on the residual test split. The "best safe fixed" threshold is selected using test labels and is therefore not available to a deployment; the bootstrap columns compare the conformal rule with a cross-validated fixed threshold over 500 resamples.

Judge	AUC	safe region	safe fraction	conformal τ	conformal dedup	best safe dedup
GPT-OSS-120B	0.995	$[0.20, 1.00]$	80%	0.80	93.9%	100.0%
GPT-OSS-20B	0.971	$[0.10, 1.00]$	90%	0.95	80.2%	97.2%
Qwen3.6-27B	0.991	$[0.05, 1.00]$	95%	0.95	90.0%	100.0%

The degradation study adds noise to calibration and test scores alike, so exchangeability is preserved by construction and the conformal guarantee is expected to hold; what the table shows is *how* it holds — the threshold rises and the layer suppresses less, rather than the error rate rising. It is a demonstration of the mechanism's failure direction, not independent evidence for the bound.

Table S18: Weak-judge transition: Gaussian noise added to the primary judge's scores (calibration and test alike, so exchangeability is preserved). As separation degrades the safe fixed-threshold region collapses, while the conformal threshold rises and keeps realised wrong suppression below α by suppressing less.

σ	AUC	safe region	safe fraction	conformal t_{hi}	conformal dedup
0.00	0.995	width 0.80	80%	0.80	93.9%
0.10	0.988	width 0.67	68%	0.74	92.1%
0.20	0.983	width 0.49	50%	0.69	84.9%
0.30	0.960	width 0.33	34%	0.76	70.0%
0.40	0.916	width 0.19	19%	0.88	50.5%
0.55	0.843	width 0.03	4%	0.99	6.8%
0.75	0.764	empty	1%	1.00	0.0%
1.05	0.690	empty	1%	1.00	0.0%
1.60	0.623	empty	1%	1.00	0.0%

S13 Schema-richness ladder

Table S19: Schema-richness ladder: one act family declared with zero, four and six typed attributes. The residual and the judge's jurisdiction shrink together. L1_loose ($n = 32$) reaches the 19 pairs needed to certify $\alpha = 0.05$ at $m = 1$, but its threshold falls at a score of 1.00, which the strict inequality never exceeds; L2_moderate ($n = 12$) and L3_strict ($n = 4$) are below it, so $t_{\text{hi}} = +\infty$. The calibrated layer therefore suppresses nothing at any level.

Level	typed attributes	refuter coverage	residual	pool n	certifiable m	t_{hi}	dedup
L1_loose	0	0.0% [0.0–3.6]	100.0% [96.4–100.0]	32	1	1.00	0.0%
L2_moderate	4	58.8% [49.1–67.9]	41.2% [32.1–50.9]	12	0	$+\infty$	0.0%
L3_strict	6	80.4% [71.6–86.9]	19.6% [13.1–28.4]	4	0	$+\infty$	0.0%

A second experiment on this ladder was intended to separate the calibration-budget component of the refinement dilemma from an intrinsically harder residual by replenishing each level's pool to a common size. It failed to do so: the replenishment pairs were drawn from a mutation class (a changed internal cost-centre allocation) that the judge treats as the same act, so adding them raised the pool's ceiling instead of its information. We report the design error rather than the comparison. The decomposition in the main paper uses a different and valid design: subsampling the declared pool to the refined pool's size.

S14 Refinement decomposition

Table S20: Refinement decomposition. The size-only counterfactual draws 5,000 random subsets of the declared pool at the refined pool's size and recomputes the threshold; "$P(\leq$ observed)" is the share of those draws whose deduplication is at most the observed refined value.

Judge	declared n	refined n	refined rank	refined dedup	size-only mean dedup	$P(\leq$ observed)
GPT-OSS-120B	142	49	2	69.0% [63.4–74.2]	88.4%	0.116
GPT-OSS-20B	142	49	2	12.1% [8.8–16.4]	55.1%	0.439
Qwen3.6-27B	139	48	2	12.1% [8.8–16.4]	57.2%	0.419

S15 Crash and paraphrased replay, per crash point

Table S21: Crash and paraphrased replay across eight crash points, 48 trials per arm, judge openai/gpt-oss-120b, paraphrase pool written by llama-3.3-70b-versatile. *Correct* means the correct final number of external effects: one wire in the duplicate arm, two in the distinct arm.

Arm	crash point	syntactic correct	semantic correct	crash point reached	window flagged
duplicate	before_fence	0/6	6/6	6/6	0/6
duplicate	after_L0	0/6	6/6	6/6	0/6
duplicate	after_witness	0/6	6/6	6/6	0/6
duplicate	after_judge	0/6	6/6	6/6	0/6
duplicate	before_external_effect	0/6	6/6	0/6	0/6
duplicate	during_external_commit_window	0/6	6/6	0/6	0/6
duplicate	after_external_effect_before_local_commit	0/6	5/6	0/6	0/6
duplicate	after_local_commit	0/6	6/6	0/6	0/6
distinct	before_fence	6/6	6/6	6/6	0/6
distinct	after_L0	6/6	6/6	6/6	0/6
distinct	after_witness	0/6	6/6	6/6	0/6
distinct	after_judge	0/6	6/6	6/6	0/6
distinct	before_external_effect	6/6	6/6	6/6	5/6
distinct	during_external_commit_window	0/6	0/6	6/6	5/6
distinct	after_external_effect_before_local_commit	0/6	0/6	6/6	6/6
distinct	after_local_commit	0/6	6/6	6/6	0/6

In the duplicate arm the window crash points are never reached, because the fence suppresses the replay before the dispatch region: a fence that recognises a duplicate does not enter the commit window on that dispatch. That is why the single-arm design used in an earlier version of this experiment could not observe the window at all, and why the distinct arm — where the fence is supposed to execute — is the one that measures it.

S16 Model panel: preregistration, results, and two defects

The study was preregistered before any judgement was collected, against a 154-item stratified sheet frozen and hashed per item. Three annotators and a blind adjudicator, none of them the primary judge, saw one pair per request with the operational definition and nothing else. All four passed the three catch trials. The channel that ran was Channel M, a panel of language models; Channel H, independent human annotators, was *not* collected, and the instrument for it ships unrun.

Defect 1: unscored items in the re-derivation 93 of the 154 sheet items are drawn from an extension corpus over nine further tools whose judge scores were never persisted. In the preregistered re-derivation those items reach the judge as abstentions, which escalate and can never be suppressed. They therefore enter the wrong-suppression denominator without being able to enter its numerator, and 12 of the 14 label flips are such items leaving that denominator. This is the mechanism behind the rise from 9.9% to 10.5% that triggered the preregistered rule. Restricted to the 51 annotated items the primary judge actually scored, wrong suppression on benign items falls from 1 of 14 to 0 of 12 under the panel's labels.

Defect 2: the published figures are not exactly reproducible The published analysis reported 116 fully annotated items. The earliest committed snapshot of the response files yields 117: one item was completed between the analysis and the commit. All 14 label flips and both re-derived rates are identical; Krippendorff's α moves from 0.828 to 0.829 and agreement from 87.9% to 88.0%.

Table S22: Model panel (Channel M), preregistered collection: agreement with the construction per stratum over 117 fully annotated items. A stratum is contested when the interval's upper bound falls below 0.80. Stratum names follow the main text; the extension-corpus attack categories are pooled with the corpus category of the same kind.

Stratum	n	agreement with construction	contested
mechanical reformat (S1)	12	100.0% [75.8–100.0]	no
paraphrase (S2)	10	100.0% [72.2–100.0]	no
loose-schema value rewrite (S3)	12	100.0% [75.8–100.0]	no
adversarial paraphrase (S4)	12	100.0% [75.8–100.0]	no
near-miss field mutation (D1)	14	92.9% [68.5–98.7]	no
legitimate repeats (D2)	12	50.0% [25.4–74.6]	yes
hand-specified semantic difference (D3)	8	87.5% [52.9–97.8]	no
typed-field injection (D4a)	12	100.0% [75.8–100.0]	no
prose-only injection (D4b)	11	45.5% [21.3–72.0]	yes
encoding-level attack	11	100.0% [74.1–100.0]	no
mis-declared field policy	3	100.0% [43.9–100.0]	no

Table S23: Re-derivation of the headline rates over the annotated subset, for each state of the response files. "As executed" is the preregistered computation, which includes items with no judge score; "scored only" restricts to the items the primary judge scored. Benign excludes the held-out injection pairs.

Collection	labels	items	WS (as executed)	dedup (as executed)	WS (scored only)	WS benign (scored only)
preregistered	constructed	117	9.9% [4.9–19.0]	78.3% [64.4–87.7]	18.2% [7.3–38.5]	7.1% [1.3–31.5]
preregistered	panel	117	10.5% [4.9–21.1]	61.7% [49.0–72.9]	15.0% [5.2–36.0]	0.0% [0.0–24.2]
head0dc8464	constructed	118	9.7% [4.8–18.7]	78.3% [64.4–87.7]	18.2% [7.3–38.5]	7.1% [1.3–31.5]
head0dc8464	panel	118	10.5% [4.9–21.1]	60.7% [48.1–71.9]	15.0% [5.2–36.0]	0.0% [0.0–24.2]
workingTree	constructed	140	10.6% [5.7–18.9]	74.5% [61.7–84.2]	20.0% [8.9–39.1]	6.7% [1.2–29.8]
workingTree	panel	140	11.9% [6.2–21.8]	57.5% [46.1–68.2]	17.4% [7.0–37.1]	0.0% [0.0–22.8]

Two further states of the response files exist, in which 1 and 23 further items reach full annotation; both are reported above as post hoc sensitivity analyses, and neither changes the direction of any finding. The provenance of all three states is recorded in `annotation/RESPONSE-PROVENANCE.md`.

What the study does and does not license A model panel controls the same threat as a human panel — whether the constructed labels are private to their author — but more weakly, because a notion of "the same act" shared by the author and the models would pass it. It validates 154 items, and at that size even a flawless result carries a 95% lower bound of 97.6%. It is never described as a human study, and the artifact's verifier fails the build if the manuscript does so.

S17 Cost

Table S24: Cost accounting, measured over each judge's final scoring session (batches of ten pairs, so the system prompt is amortised over the batch). Judge calls per 1,000 dispatches follow from the judge's jurisdiction (76.2% declared, 58.9% refined).

Judge	session pairs	tokens per comparison	mean latency per call (ms)	tokens per 1,000 (declared)	tokens per 1,000 (refined)
GPT-OSS-120B	111	183.7	2148	140,022	108,150
Qwen3.6-27B	61	513.2	6044	391,162	302,125
GPT-OSS-20B	111	199.1	1611	151,716	117,182

S18 Errata against the earlier long-form version

This work previously existed as a 44,600-word manuscript. The present paper is a rewrite, and the audit that produced it corrected the following. Each entry names what was wrong and what replaced it.

1. **Cost was understated by roughly an order of magnitude.** The cost model divided the tokens of one scoring session by every pair ever scored. Corrected: about 184 tokens per comparison and roughly 140,000 tokens per 1,000 dispatches for the primary judge, against a previously reported 5,431.
2. **Jurisdiction figures were from a superseded corpus.** 73.5%/26.5% became 76.2%/23.8%, and the refined-schema figure 56.5% became 58.9%.
3. **The conformal index at the refined pool was wrong.** At $n = 49$ and $\alpha = 0.05$ the index is 48, the second largest score, not 35 (a leftover from a 955-pair corpus in which the pool held 35 pairs).
4. **The multiplicity claim was mis-stated.** The uncorrected rule exceeds α by a factor of about eighteen at $m = 50$ (89.0% against 5%), not "a factor of nine".
5. **The refinement was described as re-declaring one tool.** It re-declares every free-text field of every tool by typing the numbers they mention.
6. **The refinement result was reported for one judge.** On the other two judges deduplication collapses to the canonical-only rate of 12.1%, which is a stronger statement of the same dilemma.
7. **The escalation-band explanation was inverted.** The earlier text attributed silent escape to saturated judges and flagged escape to graded ones; the data show the opposite pattern, and the mechanism is threshold ties (Section S8).

8. **Commit-window counts disagreed between sections.** The correct figures are 12 of 12 re-fires inside the window on the distinct arm, 11 of them flagged.
9. **Stale paired-test counts.** The deduplication comparison against the canonical baseline is 232 discordant items, not 202; the cross-judge comparison is 46/12, not 42/12.
10. **A stale layered-configuration row.** The witness-free configuration's wrong suppression is 3.4%, not 3.0% (the latter is the same configuration under a mismatched pool).
11. **Per-tool calibration and the ensembles were reported under a mismatched pool.** Under the matched pool per-tool calibration costs 27 points of deduplication, not 5, and the mean ensemble's prose-injection rate is 44.0%, not 88.0%.
12. **The relabelling thresholds were stale** ($t_{\text{hi}} = 0.7$ and 0.96; the correct values are 0.80 and 0.95).
13. **The self-recognition table was computed under a mismatched pool**, where every generator arm was deduplicated completely and no difference could show. Under the matched pool the arms differ and still show no self-recognition advantage.
14. **A rule-of-three figure was wrong** (1.5% instead of 1.3% at $n = 232$).
15. **The abstract claimed stability under "a blinded panel"** while the same paper reported that the panel's preregistered test had triggered against the safety claim. The present paper reports the trigger, its cause, and the corrected analysis.
16. **Two undisclosed provenance facts** are now disclosed: the annotated sheet draws 93 of 154 items from an unscored extension corpus, and the published panel figures are not exactly reproducible from any committed file state (Section S16).

S19 Reproducibility

Environment Node.js $\geq$ 20.11 (v24.13.0 used), no runtime dependencies; Windows 11; MiKTeX with `elsarticle` for the documents. Every analysis in this paper is offline: it reads the persisted corpus and scores and makes no model calls.

Model access Judges and generators were reached through Groq's OpenAI-compatible endpoint in August 2026 at temperature 0 (judges) and 0.9 (paraphrase generators), in batches of ten pairs per judge call. Model weights behind an endpoint are not pinned, and two generator models used for earlier arms are no longer reachable on the key, so the corpus is frozen and released rather than regenerated. A re-run against an updated endpoint is a different experiment and needs a fresh calibration.

Seeds Corpus 20260712; additive growth 20260824; annotation export and crash/replay 20260818; refinement decomposition 20260914. All resampling and permutation procedures use seeded generators recorded in the scripts.

Judge prompt The system prompt is fixed and reproduced verbatim:

```
You are an effect-identity judge inside an agent runtime.
You are shown pairs of tool calls (same tool).  For each pair decide:  do call A and call
B denote the SAME single intended real-world action -- such that executing both would
cause a duplicate effect (double charge, duplicate email, duplicate transfer)?  Or are
```

```
they TWO DISTINCT intended actions that should both execute?

Rules:
- Formatting, key order, whitespace, case of codes, number formats, spelled-out numbers,
abbreviations, aliases, and paraphrased free text do NOT make actions different.
- Different amounts, different recipients/accounts/ids, different referenced
orders/invoices, or explicit distinguishing tokens ("installment 1 of 2" vs "2 of 2")
DO make actions different.
- A retried action after a crash is the same action even if reworded.
Output STRICT JSON only:  an array with one object per pair, in order:
[{"i":  <pair index>, "p_same_pct":  <integer 0-100>}]
p_same_pct is your probability that the pair is the same single action, expressed as a
WHOLE-NUMBER PERCENTAGE.
It MUST be an integer between 0 and 100 inclusive.  Write 95 for "95% likely the same
action".
Do NOT write a decimal fraction:  0.95 is NOT accepted and will be rejected as invalid.
```

Commands The offline pipeline regenerates every number in the paper:

```
cd research/effect-identity
npm test
node src/experiments/run.js
node src/experiments/residualCalibration.js
node src/experiments/baselineMatrix.js -corpus v1
node src/experiments/attackMatrix.js -corpus v1
node src/experiments/conformalAudit.js -corpus v1
node src/experiments/crossFamily.js
node src/experiments/sensitivity.js
node src/experiments/seedSplit.js
node src/experiments/schemaLadder.js
node src/experiments/thresholdSweep.js
node src/experiments/weakJudge.js
node src/experiments/nonLlmBaseline.js
node src/experiments/aijAnalyses.js
node src/experiments/annotation.js -ingest \
-responses-dir annotation/preregistered-collection -channel M
node src/experiments/report.js > results/REPORT.md
node src/experiments/verify.js
node src/experiments/checkPaper.js
```

The steps that require model access — corpus generation, judge scoring, the multi-candidate experiment and the crash/replay experiment — are documented in `repro.config.json` with the commands, models and dates used; their outputs are shipped, so the offline pipeline above reproduces every reported number without them.

Gates `verify.js` checks eleven assertion classes against the shipped data (split separation, attack pairs excluded from calibration, score validity, abstention handling, jurisdiction-matched pools, multiplicity budgets, and that the panel study is never described as a human study). `checkPaper.js` extracts every interval quoted in the manuscript and supplement and fails unless each matches a computed rate. `tools/check-claims.mjs` checks the headline numbers quoted without intervals, rejects strings known to be stale from earlier versions, and fails on any unqualified use of “exactly-once”.

S20 Artifact contents

- `corpus/` — the 1,066-pair corpus, the judge scores for every admitted judge, the schema ladder, and the multi-candidate units.
- `src/` — canonicalisation, conformal calibration, the fence, the judge client and strict parser, the shared evaluation core, and one script per experiment.
- `results/` — every results file behind the paper, including the preserved published panel analysis and `REPORT.md`, a regenerated digest of every rate with its interval and basis.
- `annotation/` — the preregistration, the frozen sheet and its hashes, the blinded instrument, the catch trials, the collected model-panel responses in three dated states, and the Channel H protocol.
- `archive/` — superseded corpora and scores, including the parser defect of Section S3.
- `repro.config.json` — every seed, split method, threshold construction, model identifier and admission parameter, in one machine-readable place, checked by `verify.js`.

S21 Tightness computations

Theorem 2 and Corollary 2 of the main text are proved in its appendix. This section reports the computations quoted after them. They involve no model and no corpus: every number is a property of order statistics of uniform variables, computed exactly from binomial tails by `src/experiments/tightness.js`, which writes `results/tightness.json`.

The exact worst case is compared with a direct simulation of the law that attains it: a pool of n independent uniforms and candidate scores $\mathrm{frac}(V + (i-1)/m)$ for one further uniform V. The simulation is there to catch an error in the derivation, not to estimate anything. In all eight configurations it lies within two standard errors of the exact value (Table S25).

Table S25: Worst case over $\mathcal{Q}_{n,m}$ (Theorem 2) against the Bonferroni bound $m\rho/(n+1)$, at $\alpha = 0.05$. The value in parentheses is the worst case of the threshold one rank lower. The simulation column is the attaining law simulated directly (200,000 draws per row); the gap column is exact and negative entries below 10^{-12} are rounding.

Configuration	n	m	k^*	bound	worst case	gap	simulated (next rank)
declared pool, primary judge	142	1	136	4.895%	4.895%	0	4.933% (5.594%)
tightened pool, primary judge	49	1	48	4.000%	4.000%	0	3.970% (6.000%)
simulation setting, m=1	200	1	191	4.975%	4.975%	0	4.974% (5.473%)
simulation setting, m=2	200	2	196	4.975%	4.975%	0	4.938% (5.970%)
simulation setting, m=5	200	5	199	4.975%	4.975%	0	4.960% (7.463%)
simulation setting, m=10	200	10	200	4.975%	4.975%	3.1×10^{-11}	4.963% (9.950%)
guidance: 199 pairs for m=10	199	10	199	5.000%	5.000%	3.5×10^{-11}	5.074% (10.000%)
m=50, pool that certifies it	999	50	999	5.000%	5.000%	8.5×10^{-11}	4.946% (10.000%)

Beyond the calibration budget, Corollary 2(c) states that no finite scale-free threshold is valid. Table S26 gives the worst case of the most conservative such threshold for several pools. The analytic condition of the corollary leaves unchecked only pools with $0 < m/(n+1) - \alpha \le \delta_{0.05} = 1.03 \times 10^{-10}$, and at $\alpha = 1/20$ no pool with $n+1 < 4.8 \times 10^8$ can fall there, because $m/(n+1) - \alpha$ is then at least $1/(20(n+1))$. The grid check is a check of that arithmetic: it evaluates the exact worst case for the smallest violating m at every $n+1 \le 10^6$, which suffices because the worst case is non-decreasing in m.

Table S26: Beyond the budget ($m > \alpha(n+1)$): the worst case of the most conservative finite threshold, the pool maximum, at $\alpha = 0.05$. Every entry exceeds α, so no finite scale-free threshold is valid. The grid check covers every $n + 1 \leq 10^6$ at $\alpha = 0.05$ and $\alpha = 0.01$ (0 exceptions at $\alpha = 0.05$, 0 exceptions at $\alpha = 0.01$).

n	m	$\alpha(n+1)$	worst case of the pool maximum
200	11	10.05	5.473%
200	25	10.05	12.434%
200	50	10.05	24.447%
142	8	7.15	5.594%
49	3	2.50	6.000%
18	1	0.95	5.263%

S22 Action authorisation: full tables

The second instance of the recipe (main text, Section 8.10). Table S27 gives every admitted judge and configuration at both levels, each calibrated on its own jurisdiction; Table S28 gives the permit rate by category for the proposed configuration. The corpus, split and configurations are described in Section 7.7 of the main text; results are regenerated by `src/experiments/authz.js`.

Table S27: Authorisation instance, every admitted judge and configuration, calibrated on its own jurisdiction. WA is wrong authorisation over the 120 not-covered test pairs, and the res. column conditions on the residual; autonomy is the permit rate over the 100 covered test pairs; B4a and B4b are permit rates over the 40 + 40 held-out attack pairs.

Judge	α	configuration	t_{hi}	WA	WA (res.)	autonomy	B4a	B4b
GPT-OSS-120B	0.05	exact scope (no model)	—	0.0% [0.0–3.1]	0.0% [0.0–4.6]	20.0% [13.3–28.9]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
GPT-OSS-120B	0.05	raw judge, $\tau = 0.5$	—	5.0% [2.3–10.5]	7.5% [3.5–15.4]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	12.5% [5.5–26.1]
GPT-OSS-120B	0.05	calibrated, no refuter	0.30	5.0% [2.3–10.5]	7.5% [3.5–15.4]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	15.0% [7.1–29.1]
GPT-OSS-120B	0.05	certifier + refuter + calibrated	0.40	5.0% [2.3–10.5]	7.5% [3.5–15.4]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	15.0% [7.1–29.1]
GPT-OSS-120B	0.01	calibrated, no refuter	0.99	0.0% [0.0–3.1]	0.0% [0.0–4.6]	51.0% [41.3–60.6]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
GPT-OSS-120B	0.01	certifier + refuter + calibrated	$+\infty$	0.0% [0.0–3.1]	0.0% [0.0–4.6]	20.0% [13.3–28.9]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
GPT-OSS-20B	0.05	exact scope (no model)	—	0.0% [0.0–3.1]	0.0% [0.0–4.6]	20.0% [13.3–28.9]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
GPT-OSS-20B	0.05	raw judge, $\tau = 0.5$	—	1.7% [0.5–5.9]	2.5% [0.7–8.7]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	12.5% [5.5–26.1]
GPT-OSS-20B	0.05	calibrated, no refuter	0.10	3.3% [1.3–8.3]	5.0% [2.0–12.2]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	12.5% [5.5–26.1]
GPT-OSS-20B	0.05	certifier + refuter + calibrated	0.20	1.7% [0.5–5.9]	2.5% [0.7–8.7]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	12.5% [5.5–26.1]
GPT-OSS-20B	0.01	calibrated, no refuter	1.00	0.0% [0.0–3.1]	0.0% [0.0–4.6]	20.0% [13.3–28.9]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
GPT-OSS-20B	0.01	certifier + refuter + calibrated	$+\infty$	0.0% [0.0–3.1]	0.0% [0.0–4.6]	20.0% [13.3–28.9]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
Qwen3.6-27B	0.05	exact scope (no model)	—	0.0% [0.0–3.1]	0.0% [0.0–4.6]	20.0% [13.3–28.9]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
Qwen3.6-27B	0.05	raw judge, $\tau = 0.5$	—	0.8% [0.1–4.6]	1.3% [0.2–6.7]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
Qwen3.6-27B	0.05	calibrated, no refuter	0.10	0.8% [0.1–4.6]	1.3% [0.2–6.7]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
Qwen3.6-27B	0.05	certifier + refuter + calibrated	0.10	0.8% [0.1–4.6]	1.3% [0.2–6.7]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
Qwen3.6-27B	0.01	calibrated, no refuter	0.90	0.8% [0.1–4.6]	1.3% [0.2–6.7]	100.0% [96.3–100.0]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
Qwen3.6-27B	0.01	certifier + refuter + calibrated	$+\infty$	0.0% [0.0–3.1]	0.0% [0.0–4.6]	20.0% [13.3–28.9]	0.0% [0.0–8.8]	0.0% [0.0–8.8]

S23 Class-conditional validity

The analysis behind Section 8.3 of the main text, computed offline from the persisted scores by `src/experiments/classConditional.js`, which writes `results/class_conditional.json` and first asserts that it reproduces the deployed configuration's headline rates.

The three benign kinds of difference in the corpus (D1 near-miss mutations, D2 legitimate repeats, D3 hand-specified semantic differences) partition the distinct pairs of the judge's jurisdiction. For each judge, Table S29 compares each category's wrong suppression under the deployed marginal threshold with its wrong suppression under a Mondrian threshold computed on that category's resid-

Table S28: Authorisation instance, permit rate by category on the test split for the proposed configuration at $\alpha = 0.05$. Categories A are covered (permitting is correct); categories B are not (permitting is wrong authorisation).

Category	GPT-OSS-120B	GPT-OSS-20B	Qwen3.6-27B
A1 whitespace (certified)	100.0% [83.9–100.0]	100.0% [83.9–100.0]	100.0% [83.9–100.0]
A2 paraphrase	100.0% [94.0–100.0]	100.0% [94.0–100.0]	100.0% [94.0–100.0]
A3 specific instance	100.0% [83.9–100.0]	100.0% [83.9–100.0]	100.0% [83.9–100.0]
B1 typed violation (refuted)	0.0% [0.0–8.8]	0.0% [0.0–8.8]	0.0% [0.0–8.8]
B2 out of scope	7.5% [3.5–15.4]	2.5% [0.7–8.7]	1.3% [0.2–6.7]

ual calibration pool alone (Corollary 1). The category is defined by the label and is not observable when the fence decides, so the per-category threshold is an oracle; the rule that is valid for every category without knowing which applies is the maximum of the per-category thresholds, and Table S30 gives its deduplication, together with Mondrian calibration by tool, a partition the fence can observe. Table S31 reweights the residual distinct test mix toward D3.

Table S29: Class-conditional validity at $\alpha = 0.05$. For each judge, the first row is the deployed (marginal) rule; each category row gives that category's residual calibration pool, its certifiable m, its Mondrian threshold t_c, and its wrong suppression on the test split under the marginal threshold and under t_c. The category is defined by the label, so t_c is an oracle; the deployable rule valid for every category is $\max_c t_c$ (last row).

Judge	population	pool n	cert. m	threshold	WS, marginal t_{hi}	WS, Mondrian / class-valid
GPT-OSS-120B	all distinct (marginal)	142	7	0.80	1.7% [0.7–4.3]	—
	D1 near-miss mutation	37	1	0.80	0.8% [0.1–4.4]	0.8% [0.1–4.4]
	D2 legitimate repeats	75	3	0.85	1.3% [0.2–7.2]	1.3% [0.2–7.2]
	D3 semantic difference	30	1	0.90	6.5% [1.8–20.7]	0.0% [0.0–11.0]
	class-valid rule ($\max_c t_c$)	—	—	0.90	—	0.0% [0.0–1.6]
GPT-OSS-20B	all distinct (marginal)	142	7	0.95	1.3% [0.4–3.7]	—
	D1 near-miss mutation	37	1	0.10	0.0% [0.0–3.0]	0.0% [0.0–3.0]
	D2 legitimate repeats	75	3	1.00	4.0% [1.4–11.1]	0.0% [0.0–4.9]
	D3 semantic difference	30	1	1.00	0.0% [0.0–11.0]	0.0% [0.0–11.0]
	class-valid rule ($\max_c t_c$)	—	—	1.00	—	0.0% [0.0–1.6]
Qwen3.6-27B	all distinct (marginal)	139	7	0.95	0.9% [0.2–3.1]	—
	D1 near-miss mutation	37	1	1.00	0.0% [0.0–3.0]	0.0% [0.0–3.0]
	D2 legitimate repeats	72	3	1.00	2.7% [0.7–9.2]	0.0% [0.0–4.9]
	D3 semantic difference	30	1	0.00	0.0% [0.0–11.0]	0.0% [0.0–11.0]
	class-valid rule ($\max_c t_c$)	—	—	1.00	—	0.0% [0.0–1.6]

Table S30: The price of per-class validity. Deduplication under the marginal rule, under the class-valid rule $\max_c t_c$ (with the category that sets it), and under Mondrian calibration by tool, an observable partition; the last column counts tools whose residual pool is below the 19 pairs needed at $m = 1$.

Judge	dedup, marginal	dedup, class-valid	binding category	WS, by tool	dedup, by tool	tools off
GPT-OSS-120B	94.7% [91.4–96.7]	69.0% [63.4–74.2]	D3 semantic difference	0.4% [0.1–2.4]	67.6% [61.9–72.8]	2
GPT-OSS-20B	82.6% [77.7–86.6]	12.1% [8.8–16.4]	D3 semantic difference	0.0% [0.0–1.6]	32.0% [26.8–37.7]	2
Qwen3.6-27B	89.0% [84.8–92.1]	12.1% [8.8–16.4]	D2 legitimate repeats	0.4% [0.1–2.4]	55.5% [49.7–61.2]	2

Table S31: Residual wrong suppression when the distinct residual test mix (144 pairs) is reweighted toward the hand-specified semantic-difference category (D3); the remaining weight is split between D1 and D2 in their observed ratio. The first column is the observed mix. Intervals are stratified percentile bootstraps; a row with no suppression in any stratum has no interval.

Judge	rule	D3 20.8% (obs.)	D3 13%	D3 25%	D3 50%
GPT-OSS-120B	marginal	2.8% [0.7–5.6]	2.4% [0.4–5.1]	3.0% [0.7–6.1]	4.2% [0.4–9.3]
GPT-OSS-120B	class-valid	0.0%	0.0%	0.0%	0.0%
GPT-OSS-20B	marginal	2.1% [0.0–4.9]	2.3% [0.0–5.3]	2.0% [0.0–4.6]	1.3% [0.0–3.1]
GPT-OSS-20B	class-valid	0.0%	0.0%	0.0%	0.0%
Qwen3.6-27B	marginal	1.4% [0.0–3.5]	1.5% [0.0–3.8]	1.3% [0.0–3.3]	0.9% [0.0–2.2]
Qwen3.6-27B	class-valid	0.0%	0.0%	0.0%	0.0%